Physiology: Airways and Pulmonary Blood Flow

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Objectives

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- Align major respiratory airway structures with their basic composition & function.
- Describe the anatomy of the conducting and respiratory parts of the respiratory tree and explain underlying structure-function relationships.
- Describe where and how autonomic air flow regulation occurs.
- Describe how socio-economic determinants of health can affect someone's ability to breathe.
- Describe the basic organization of an alveolus and assign functions to the cell types it contains.
- Explain the relationship between the surface tension of water and surfactant on one side, and alveolar collapse and the Laplace Law on the other.
- Compare and contrast basic setup and function of pulmonary and bronchial circulation.
- Project the effect of socio-economic determinants of health on common respiratory system pathologies.

Respiration –External & Internal

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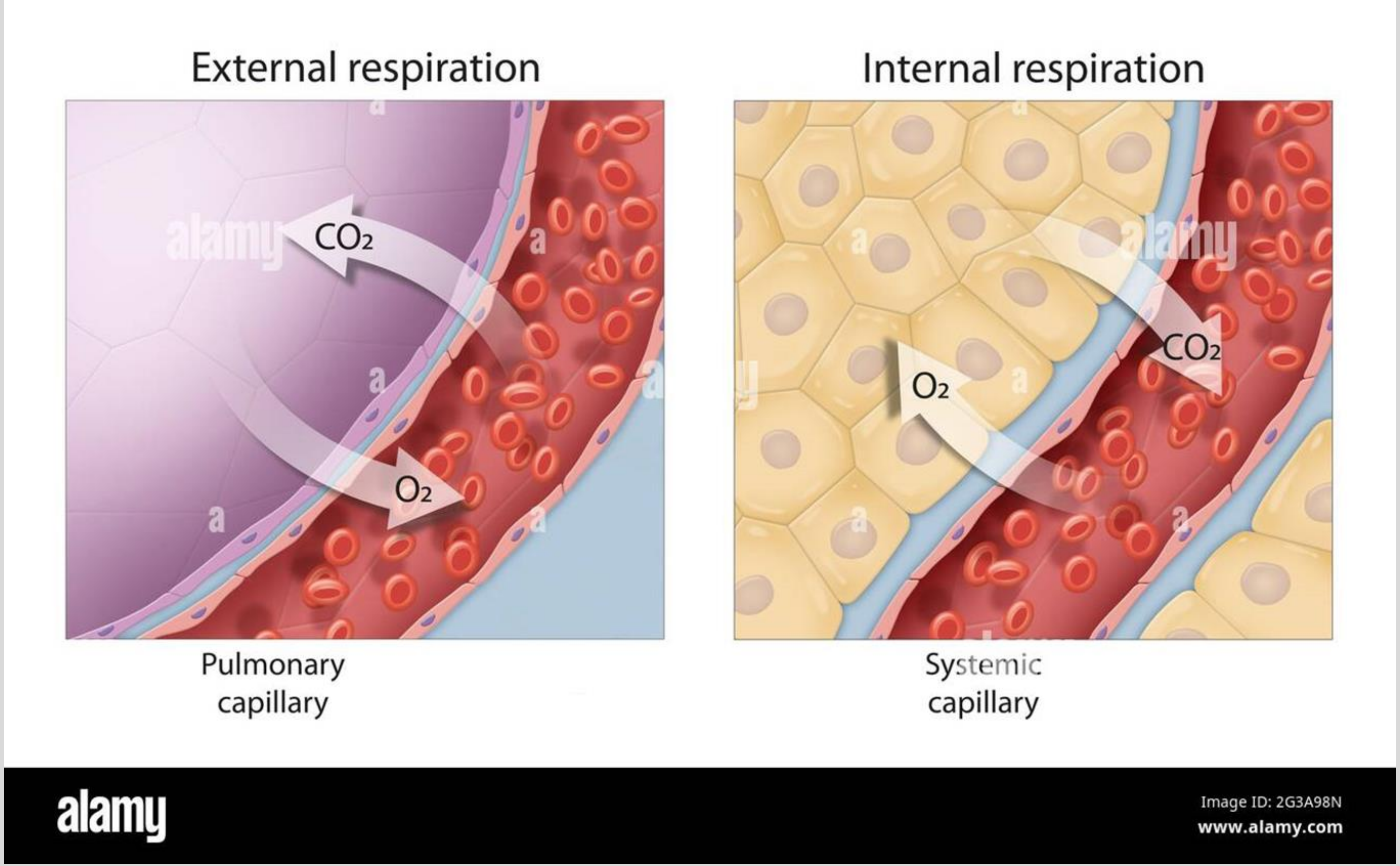
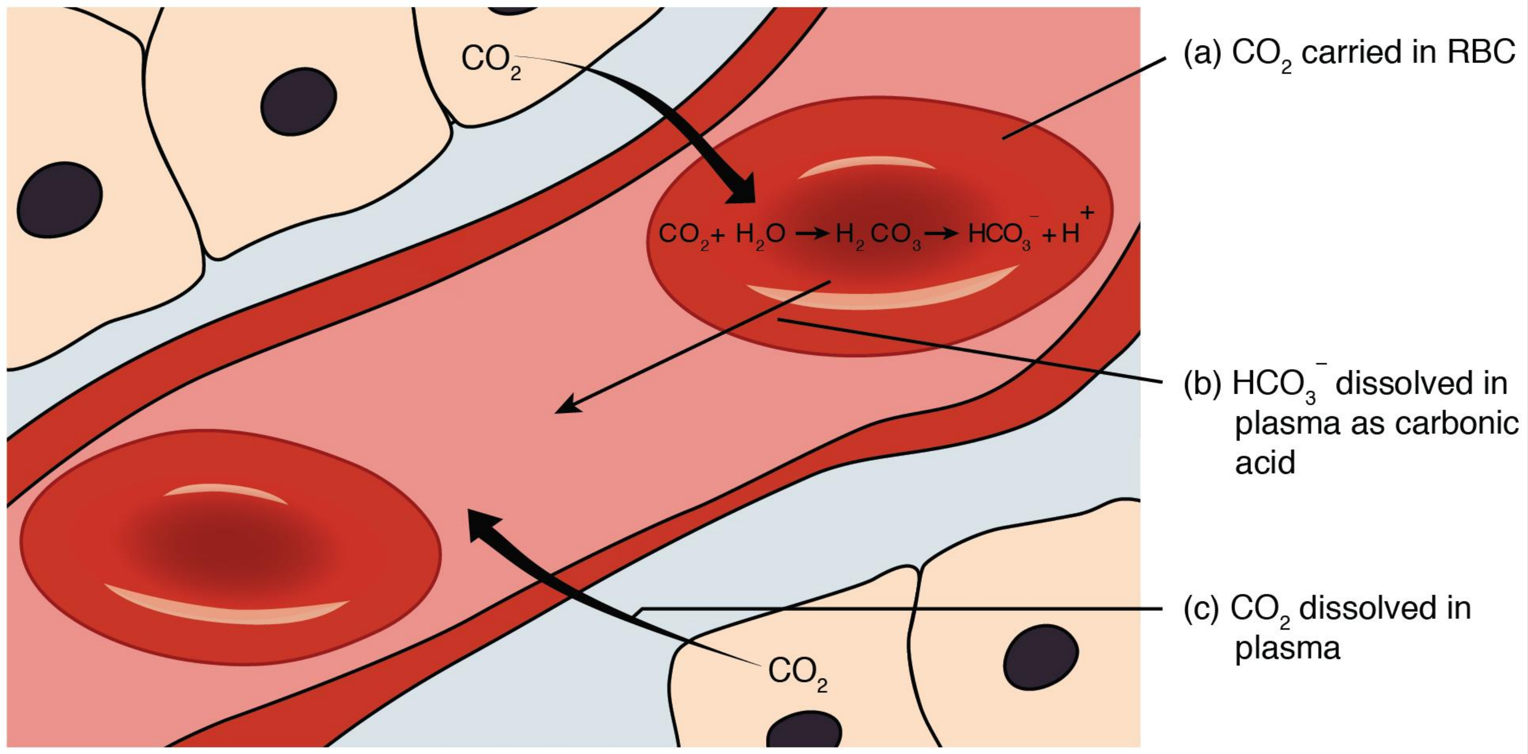


CO_2 & the Bicarbonate Buffer System



Respiratory Tree

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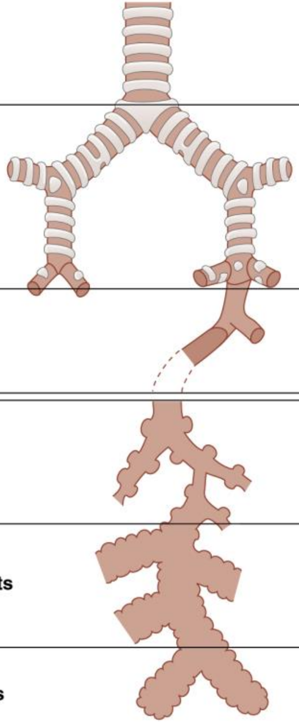


Zone	Structure	Number	Cilia	Smooth Muscle	Cartilage
CONDUCTING ZONE	Trachea	1	Yes	Yes	Yes
	Bronchi	2 4 8	Yes	Yes	Patchy
	Bronchioles	–	Yes	Yes	No
RESPIRATORY ZONE	Respiratory bronchioles	–	Some	Some	No
	Alveolar ducts	–	No	Some	No
	Alveolar sacs	6×10^8	No	No	No

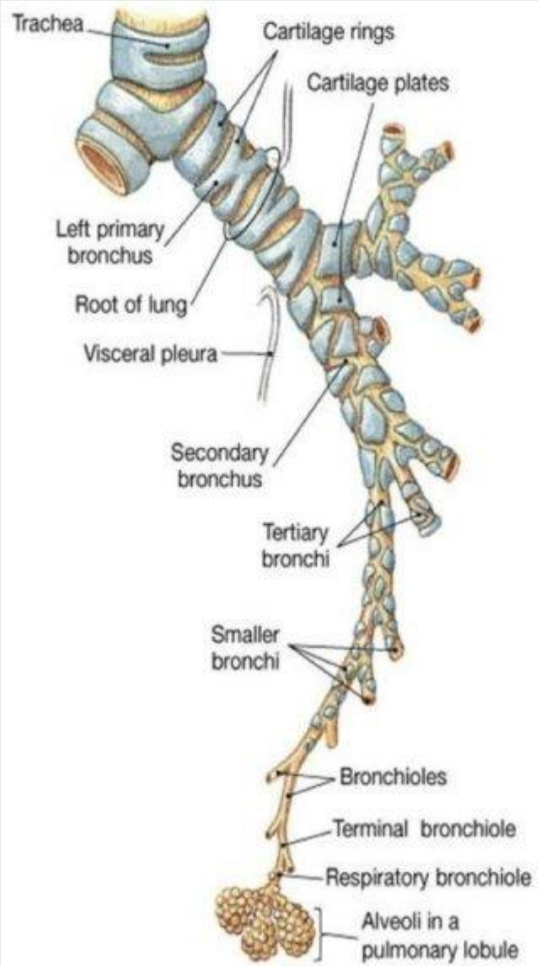
Generation 0

Level of 3rd or 4th thoracic vertebra

Generation 10: Cartilage Stops

Generation 23

Cartilage



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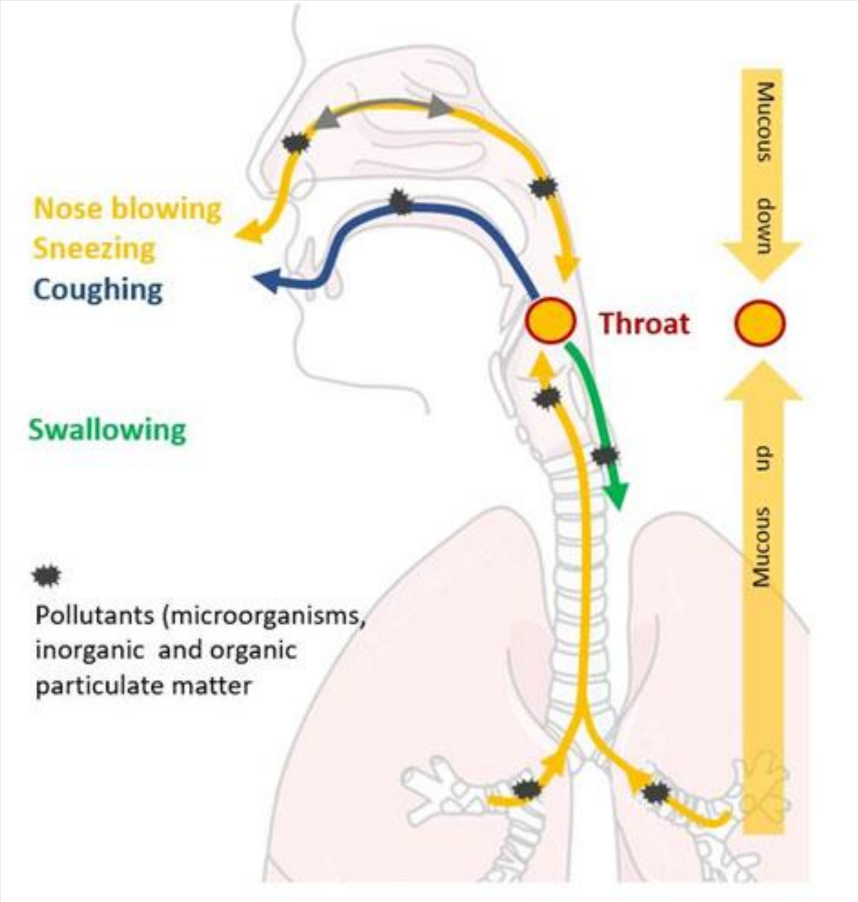
Conducting Zone – Mucociliary Clearance

Aka Mucociliary Elevator



- Humidify
- Warm
- Filter

Autonomic Regulation of Respiratory Smooth Muscle

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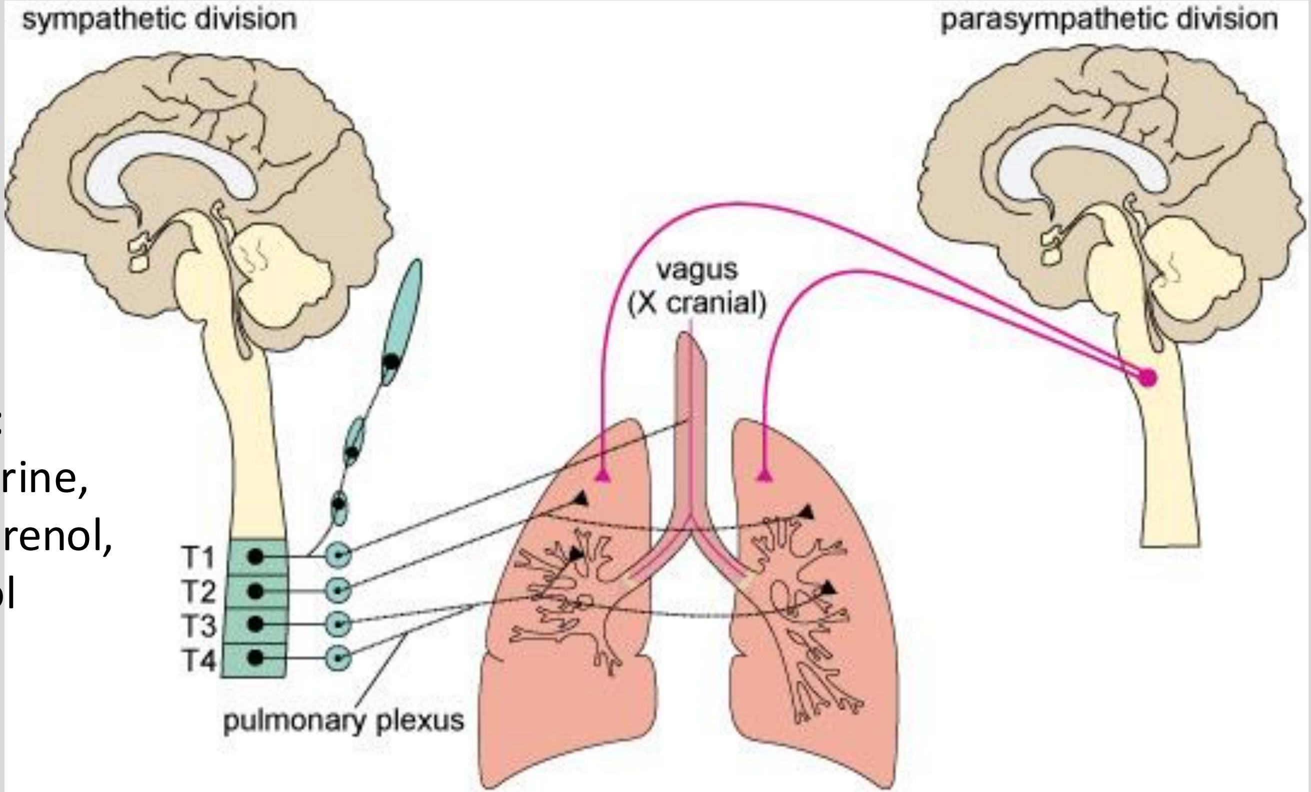
Dilate with:

- epinephrine,
- isoproterenol,
- albuterol



Beta-2 adrenergic receptors

Muscarinic Ach receptors

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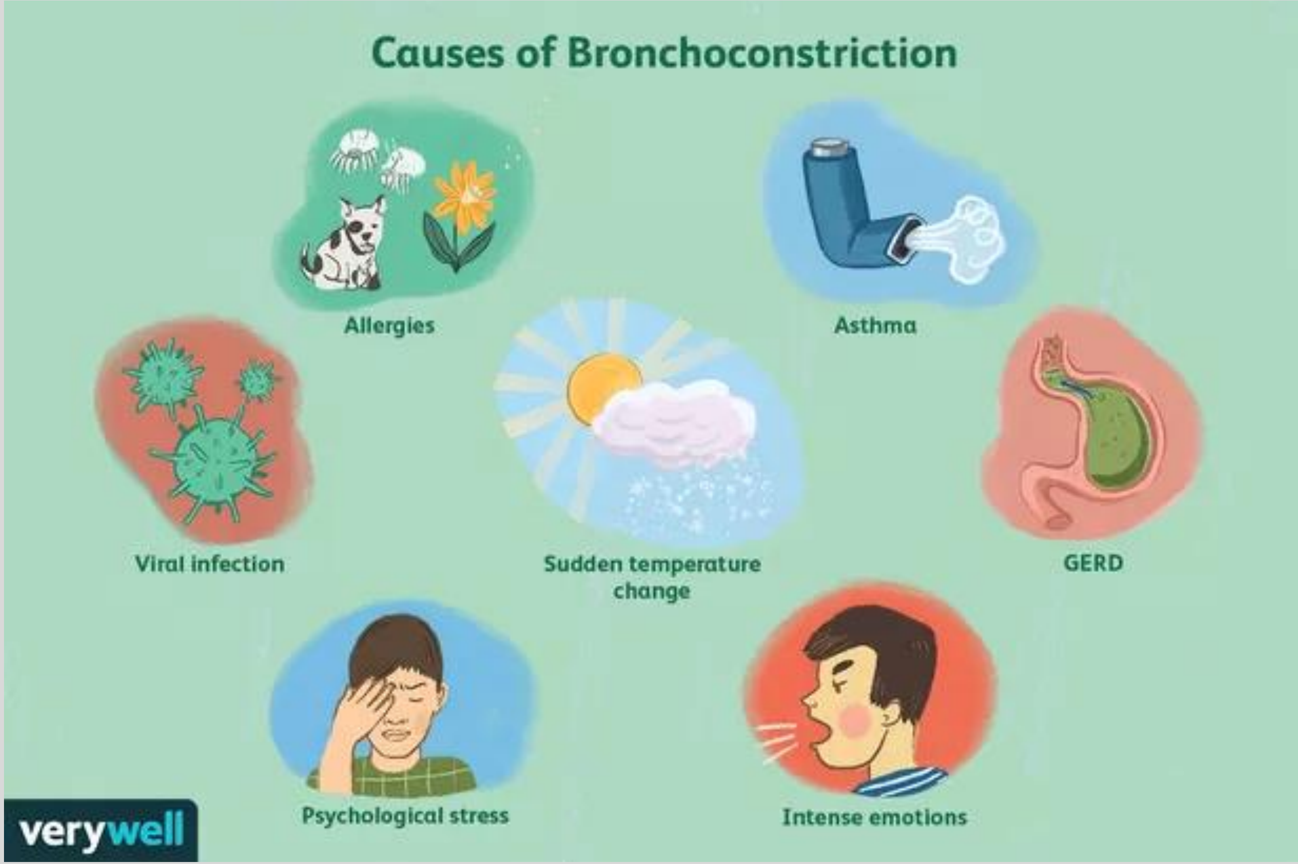

Socio-economic Determinants of Health

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A Case of Asthma and Allergic Rhinitis

Dylan is a 7-year-old African American male presenting to the emergency department of the New York-Presbyterian/Columbia University Irving Medical Center with severe shortness of breath, chest pain, and a wheezing cough. Dylan's mother is crying and afraid that he could asphyxiate. The treating physician manages to calm her down and notes that Dylan has been suffering from similar but less severe symptoms, including episodic cough, wheeze, shortness of breath and chest tightness with itchy red watery eyes and a stuffy, runny, nose, for more than a year. When asked about their living circumstances, Dylan's mother explains that she, her son and 1 younger sibling live in a small 1-bedroom apartment in Washington Heights, Manhattan, NY where she sleeps on the couch. She is working at a local bakery around the corner but recently had to reduce her hours due to issues with childcare. She also states that she is behind with the rent and that her landlord is ignoring her plea to fix or replace her window air conditioning unit subjecting her and the children to hot, low-quality air during the summer months, likely triggering Dylan's asthma attack.

Case author: JR Leheste

Respiratory Zone – Gas Exchange

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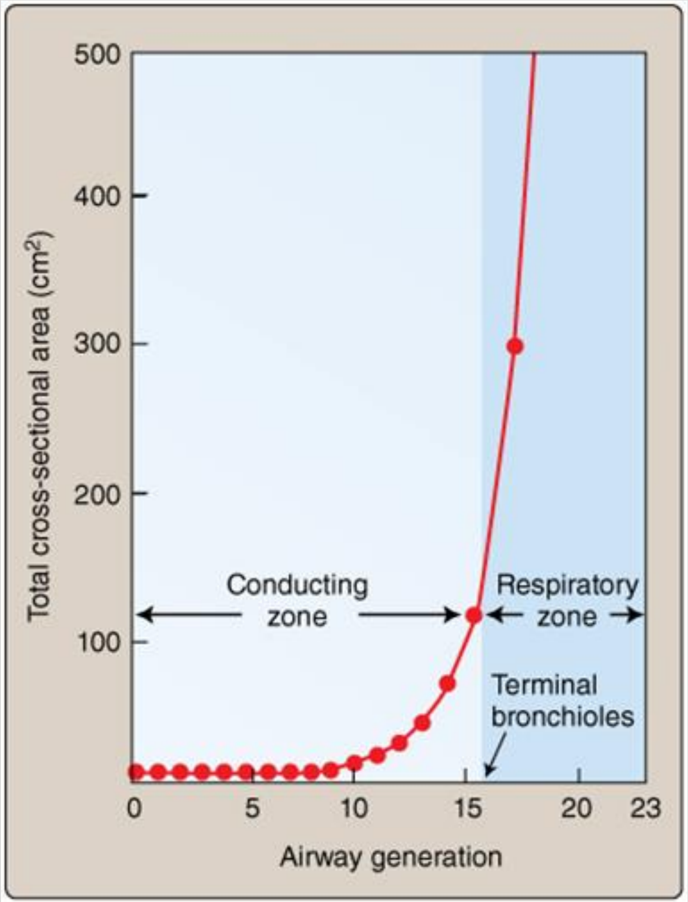
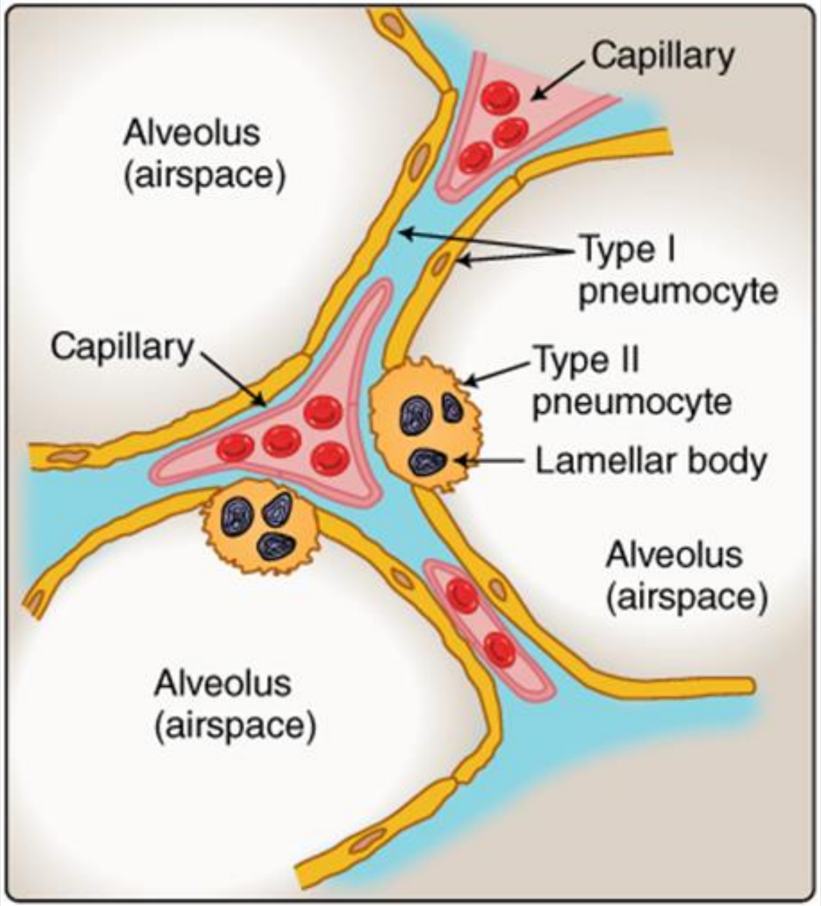


300 Million Alveoli per Lung



~ 90%

~ 10% → Surfactant, stem cell

Pulmonary Circulation

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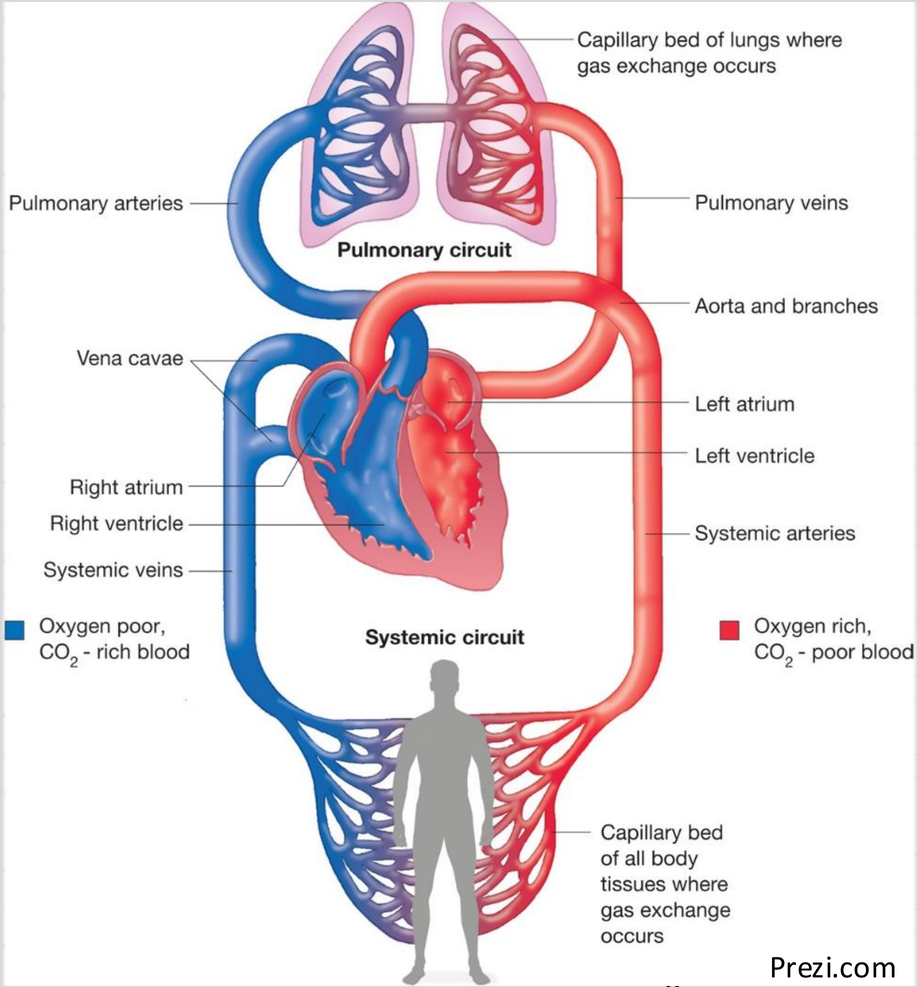

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Pulmonary Circulation Continued

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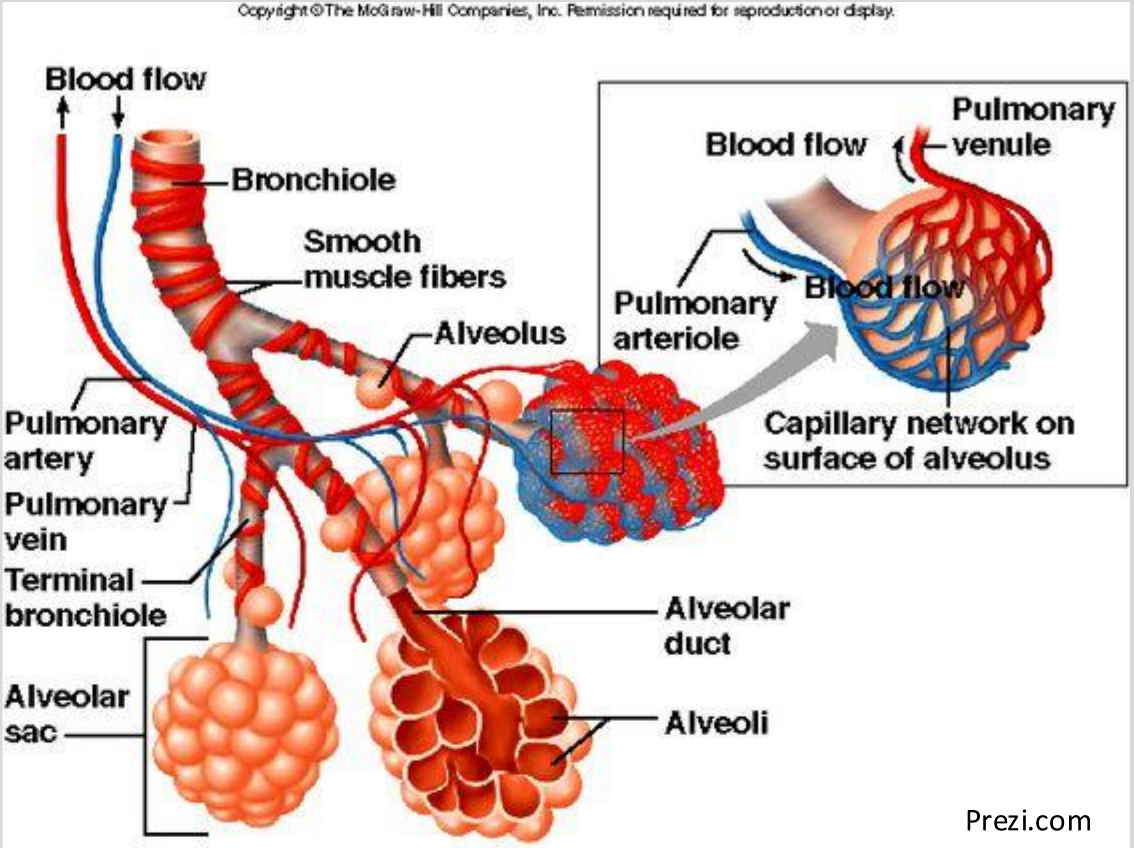

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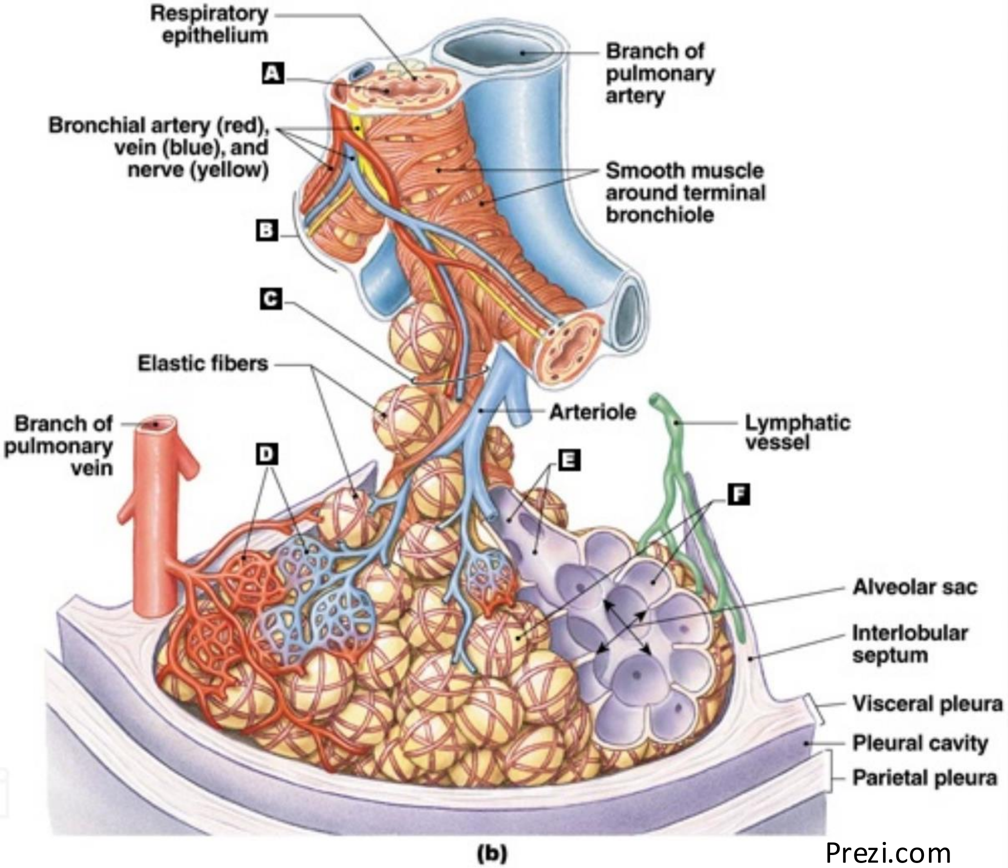
Bronchial Circulation



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Heads-Up on Shunts

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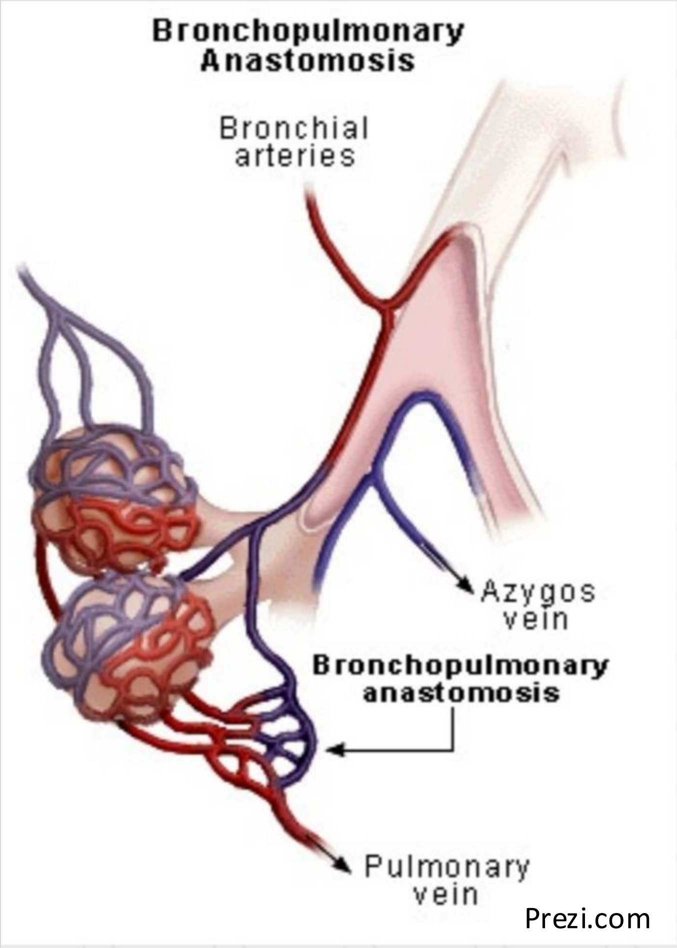

2/3 drain here

Alveolar Lining Fluid & Surface Tension

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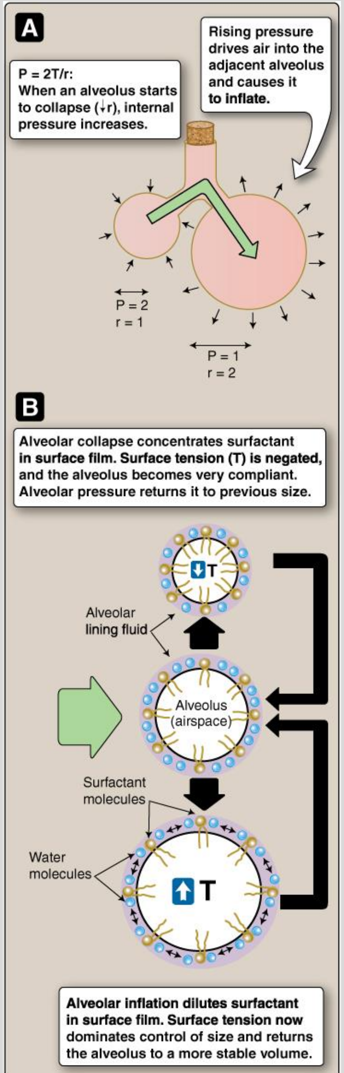
https://gifer.com/en/HUPD

Laplace Law

The pressure inside an inflated elastic container with a curved surface, e.g., a bubble or a blood vessel, is **inversely proportional to the radius** as long as the surface tension is presumed to change little



→ Increased compliance

→ Prevents over-wetting

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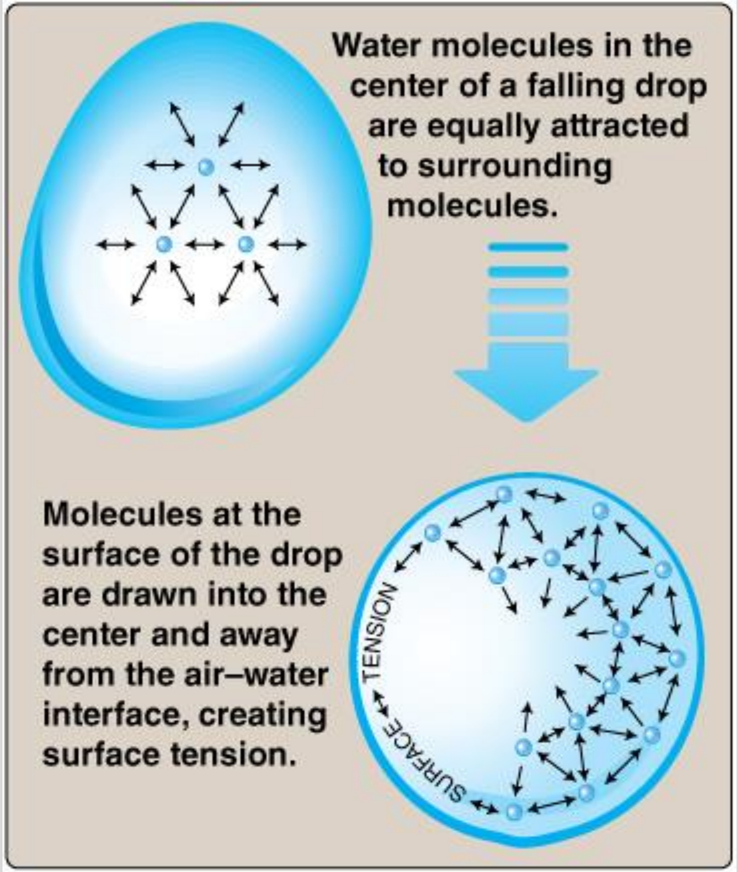

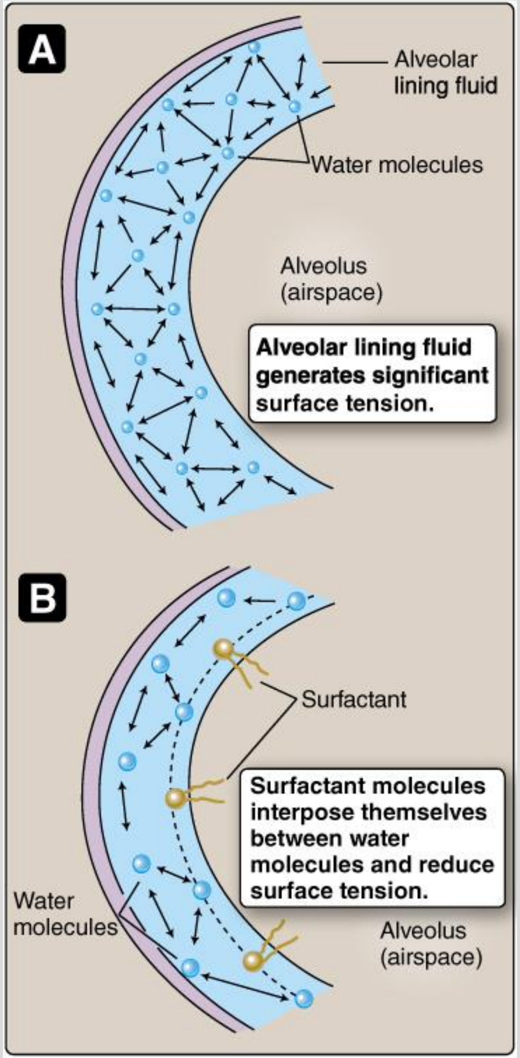

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Dipalmitoyl phosphatidylcholine (DPPC), a phospholipid

Practice Question

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A team of biomedical researchers is investigating lung physiology using an animal model. They temporarily occlude the pulmonary artery supplying the left lung while leaving the bronchial arteries intact. They observe that while gas exchange in the left lung drops to near zero, the structural tissues of the left conducting airways remain healthy and viable. Which of the following concepts best explains why the tissues of the conducting airways survived despite the occlusion of the pulmonary artery?

A. The pulmonary artery normally supplies oxygen-rich blood to the conducting airways, but the bronchial arteries quickly reversed their flow to compensate.

B. The bronchial circulation branches from the systemic arterial tree to provide oxygenated blood directly to the structural walls of the conducting airways.

C. The pulmonary circulation is a high-pressure system meant for tissue nutrition, while the bronchial circulation is exclusively used for alveolar gas exchange.

D. The alveolar capillary beds supplied by the pulmonary artery provide the primary oxygen source for the thick walls of the main and segmental bronchi.

E. Both the pulmonary and bronchial arterial systems originate directly from the right ventricle, allowing total redundant backup of nutrient delivery.

Resources

Updated resources can be found in the Class Preparation Guide (CPG) that goes with this lecture.

Lecture Feedback Form

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R